

- Description

This module manages a smart transformer current monitoring system, ensuring that:

1. **The electrical current passing through the transformer is continuously measured and compared against safe limits.**
2. **If the current exceeds the maximum safe threshold, an overload warning is triggered.**
3. **If the current drops too low, an underload alert is issued, indicating possible faults or disconnections.**
4. **A log of all current readings, warnings, and corrective actions is maintained for system reliability and optimization.**

This system helps prevent transformer failures, improve efficiency, and ensure a stable power supply.

- Specification

module SmartTransformerMonitoring

var currentLoad, maxCurrent, minCurrent: real

var log: sequence of string

process updateCurrent(changeAmount: real)

functional scenario form:

currentLoad ≤ maxCurrent && log = log ⊕ ("Current Load Updated: " + currentLoad + " A"), Current load updated successfully.

||

currentLoad > maxCurrent && log = log ⊕ ("Overload Warning: " + currentLoad + " A, Risk of Transformer Damage."), Overload warning triggered, transformer at risk.

||

currentLoad < minCurrent && log = log ⊕ ("Underload Alert: " + currentLoad + " A, Possible Fault or Disconnection."), Underload alert triggered, possible fault detected.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Transformer {
    private double maxCurrent;
    private double minCurrent;
    private double currentLoad;
    private List<String> log;
    public Transformer(double maxCurrent, double minCurrent, double initialLoad) {
        this.maxCurrent = maxCurrent;
        this.minCurrent = minCurrent;
    }
}
```

```

        this.currentLoad = initialLoad;
        this.log = new ArrayList<>();
    }
    public void updateCurrent(double changeAmount) {
        currentLoad += changeAmount;
        String message = String.format("Current Load: %.2f A", currentL
oad);
        if (currentLoad > maxCurrent) {
            message += " | WARNING: Overload Detected! Risk of Transfor
mer Damage.";
        } else if (currentLoad < minCurrent) {
            message += " | ALERT: Underload Detected! Possible Fault or
Disconnection.";
        }
        System.out.println(message);
        log.add(message);
    }
    public void printLog() {
        System.out.println("\nTransformer Load Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c37_TransformerMonitoringSystem {
    public static void main(String[] args) {
        Transformer transformer = new Transformer(200.0, 50.0, 100.0);
        // testcase-VT:
        //     transformer.updateCurrent(80.0);
        //     transformer.updateCurrent(-60.0);
        //     transformer.updateCurrent(-80.0);
        //     transformer.updateCurrent(90.0);
        //     transformer.updateCurrent(90.0);
        //     transformer.printLog();
        //
        // testcase-FT:
        //     transformer.updateCurrent(4.185580495846825E298);
        //     transformer.updateCurrent(1.7976931348623157E308);
        //     transformer.updateCurrent(-1.7976931348623157E308);
        //     transformer.updateCurrent(-4.9E-324);
        //     transformer.updateCurrent(-1.7976931348623157E308);
        //     transformer.printLog();
        //
        // testcase-Z3:

```

```
//      transformer.updateCurrent(8);
//      transformer.updateCurrent(6);
//      transformer.updateCurrent(8);
//      transformer.updateCurrent(9);
//      transformer.updateCurrent(10);
//      transformer.printLog();

// testcase-UVT:
transformer.updateCurrent(80);
transformer.updateCurrent(-60);
transformer.updateCurrent(80);
transformer.updateCurrent(90);
transformer.updateCurrent(90);
//      transformer.printLog();
}
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	838	844
▼ src	0.7 %	6	838	844
▼ code	0.7 %	6	838	844
▼ c37_TransformerMonitoringSystem.java	100.0 %	6	0	6
> c37_TransformerMonitoringSystem		0	0	0
> Transformer	100.0 %	6	0	6

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	106	20,539	20,645
▼ src	0.5 %	106	20,539	20,645
▼ code	0.5 %	106	20,539	20,645
▼ c37_TransformerMonitoringSystem.java	97.2 %	106	3	109
> c37_TransformerMonitoringSystem	89.3 %	25	3	28
> Transformer	100.0 %	81	0	81

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	841	844
▼ src	0.4 %	3	841	844
▼ code	0.4 %	3	841	844
▼ c37_TransformerMonitoringSystem.java	50.0 %	3	3	6
> c37_TransformerMonitoringSystem		0	0	0
> Transformer	50.0 %	3	3	6

5. Fuzz testing instruction coverage: 89%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	97	20,548	20,645
▼ src	0.5 %	97	20,548	20,645
▼ code	0.5 %	97	20,548	20,645
▼ c37_TransformerMonitoringSystem.java	89.0 %	97	12	109
> c37_TransformerMonitoringSystem	89.3 %	25	3	28
> Transformer	88.9 %	72	9	81

6. Z3 branch coverage: 33.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	838	840
▼ src	0.2 %	2	838	840
▼ code	0.2 %	2	838	840
▼ c37_TransformerMonitoringSystem.java	33.3 %	2	4	6
> Transformer	33.3 %	2	4	6
> c37_TransformerMonitoringSystem		0	0	0

7. Z3 instruction coverage: 72.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	78	22,306	22,384
▼ src	0.3 %	78	22,306	22,384
▼ code	0.3 %	78	22,306	22,384
▼ c37_TransformerMonitoringSystem.java	72.9 %	78	29	107
> Transformer	67.9 %	55	26	81
> c37_TransformerMonitoringSystem	88.5 %	23	3	26

8. Unimproved V-method branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	837	840
▼ src	0.4 %	3	837	840
▼ code	0.4 %	3	837	840
▼ c37_TransformerMonitoringSystem.java	50.0 %	3	3	6
> Transformer	50.0 %	3	3	6
> c37_TransformerMonitoringSystem		0	0	0

9. Unimproved V-method instruction coverage: 76.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	82	22,302	22,384
▼ src	0.4 %	82	22,302	22,384
▼ code	0.4 %	82	22,302	22,384
▼ c37_TransformerMonitoringSystem.java	76.6 %	82	25	107
> Transformer	72.8 %	59	22	81
> c37_TransformerMonitoringSystem	88.5 %	23	3	26